

Homework 8

UT EID: kt27955 Github Link: <https://github.com/ktaspa/Homework8SDS315>

Problem 1: Regression Warm-Up

A)

We fit a linear regression model with creatclear as the response and age as the predictor. Then with the model we get the value 113.723 for the creatinine clearance rate we expect for a 55-year-old.

```
##          1
## 113.723
```

B)

The rate of change of creatinine clearance is negative so it decreases with age. We get the slope of the linear regression.

```
## [1] -0.6198159
```

C)

Compared the actual creatinine clearance rates of two patients to the expected values from the model and got the 40-year-old with a rate of 135 having a higher creatinine clearance rate. Equation: observed - predicted is difference between the two groups.

```
##          1          1
## 11.979718  1.376035
```

Problem 2: Modeling Disease Growth

1.

The growth rate is 0.183 and the doubling time is 3.8.

The growth rate confidence interval with 95% confidence is (0.162, 0.204)

The doubling time confidence interval with 95% confidence is (3.4, 4.3)

Both intervals are positive and go not contain 0, meaning that both values are statistically significant.

```
## days_since_first_death
##          0.183218

## days_since_first_death
##          3.783183

##    2.5 %    97.5 %
## 0.1623227 0.2041133

##    97.5 %    2.5 %
## 3.395895 4.270180
```

2.

The growth rate is 0.276 and the doubling time is 2.5.

The growth rate confidence interval with 95% confidence is (0.237, 0.316)

The doubling time confidence interval with 95% confidence is (2.2, 2.9)

Both intervals are positive and do not contain 0, meaning that both values are statistically significant.

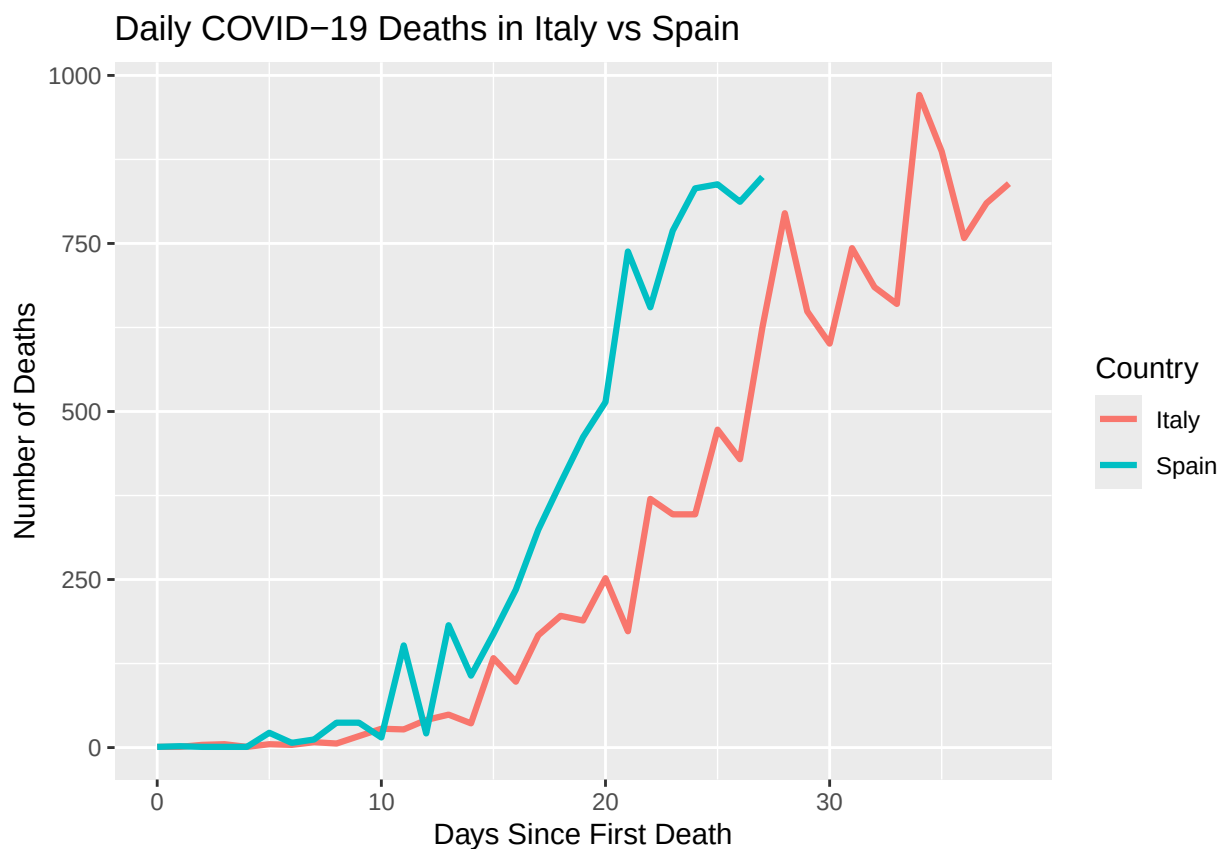
```
## days_since_first_death
##          0.2762447

## days_since_first_death
##          2.509178

##      2.5 %      97.5 %
## 0.2369611 0.3155283

##      97.5 %      2.5 %
## 2.196783 2.925151
```

3.



Problem 3: Price Elasticity of Demand

The price elasticity of milk is -1.619. To estimate the elasticity I used a model that compares the percent change in sales to the percent change in price and the slope of that line tells us the price elasticity.

```
## log_price
## -1.618578
```

Confidence interval done with confint at the 95% confidence level gave an interval of (-1.779, -1.458) the

interval is all negative and does not contain 0 which means that the price lelasticity of milk is statistically significant.

```
##      2.5 %    97.5 %  
## -1.779358 -1.457798
```